

Nature Portfolio Reporting Summary — content for the official form

“Same Prompt, Different Answer: Hidden Non-Determinism in LLM APIs Undermines Scientific Reproducibility”

Content to transcribe into the official Nature Portfolio Reporting Summary PDF (XFA form). Corresponding author: Lucas Rover (ORCID: 0000-0001-6641-9224) Coauthors: Hugo Valadares Siqueira (0000-0002-1278-4602); Eduardo Tadeu Bacalhau (0000-0002-3936-0375); Anibal Tavares de Azevedo (0000-0003-1678-7795); Yara de Souza Tadano (0000-0002-3975-3419)

1. Statistics

For all statistical analyses, confirm that the following items are present in the figure legend, table legend, main text, or Methods section.

#	Item	Status	Where in manuscript
1	Exact sample size (n) for each group	Confirmed	Table 1 (main text): n per deployment stack per task. Supplementary Table S7: full coverage matrix (e.g., 30 abstracts $\times$ 5 reps = 150 runs). Total: 7,004 runs. Methods §Tasks and §Input data.
2	Statement on distinct vs repeated samples	Confirmed	Methods §Experimental conditions: 5 repetitions of the same prompt per item (repeated measures). Each abstract / HumanEval problem / GSM8K problem is a distinct input.
3	Statistical test(s) used, one- or two-sided	Confirmed	Methods §Statistical analysis: Fisher’s exact test (two-sided), Mann–Whitney U (two-sided), Wilcoxon signed-rank (two-sided), bootstrap CIs (10,000 resamples, percentile method). Holm–Bonferroni correction across 68 tests. All tests two-sided.
4	Description of all covariates tested	Confirmed	Methods §Experimental conditions: temperature (0.0, 0.3, 0.7), seed (fixed vs variable), deployment mode (local vs API), task type, prompt format (chat vs completion). <i>Deployment mode</i> refers to whether a stack is served locally (single-GPU Ollama on Apple M4) or via a remote cloud API; it is a stack-level covariate fixed for the lifetime of each experimental group, not a per-run varying parameter. It is distinct from <i>decoding mode</i> (greedy at t=0, fixed across deterministic conditions C1–C2) and from <i>prompt format</i> (controlled in Supplementary §S7 with the chat-format control experiment).
5	Assumptions or corrections	Confirmed	Methods §Statistical analysis: Holm–Bonferroni correction for 68 multiple comparisons. Non-parametric tests chosen (no normality assumption). Bootstrap percentile method for CIs. Supplementary §S9 details the full correction procedure.
6	Central tendency and variation/uncertainty	Confirmed	All EMR values reported with 95% bootstrap CIs in brackets (e.g., 0.443 [0.32, 0.57]). Table 1, Extended Data Tables 1–7.
7	Test statistic, CI, effect sizes, df, P values	Confirmed	Cliff’s 0.784–0.896 (large to very large); Cohen’s d > 1.6 (paired); Cohen’s h 0.40–3.14. Per-field paired Cohen’s d = +1.41 (conclusion-relevant vs metadata fields). P values exact where possible. Supplementary §S9 (Holm–Bonferroni results for all 68 tests).
8	Bayesian analysis priors/MCMC	n/a	No Bayesian analysis performed.

#	Item	Status	Where in manuscript
9	Hierarchical/complex designs	Confirmed	No hierarchical models. Bootstrap CIs computed at the per-item level, then aggregated.
10	Effect size estimates		Cliff's (non-parametric) for local vs API: 0.784–0.896. Cohen's d > 1.6 (paired). Cohen's h 0.40–3.14 (Fisher). Per-field Cohen's d = +1.41. Reported in Results and Supplementary §S9.

2. Software and code

Data collection. All experiments conducted using custom Python scripts (Python 3.14.3). Local stacks: Ollama v0.15.5 serving LLaMA 3 8B, Mistral 7B, Gemma 2 9B. API stacks: OpenAI Python SDK v1.59.9 (GPT-4 gpt-4-0613 and gpt-4o-2024-11-20); urllib-based runners for Anthropic (Claude Sonnet 4.5), Google (Gemini 2.5 Pro), DeepSeek Chat, Perplexity Sonar, Together AI (LLaMA 3 8B INT4 quasi-isolation probe). Loaders for HumanEval and GSM8K via standard benchmark formats. All API payloads documented in Supplementary §S4. Data collection scripts available in the project repository.

Data analysis. Python 3.14.3 with: scipy (statistical tests), scikit-learn (metrics), rouge-score (ROUGE-L), bert-score (BERTScore F1, roberta-large), matplotlib and seaborn (figures), json and hashlib (provenance hashing). Bootstrap CIs computed with custom script (10,000 resamples, percentile method). Two-judge LLM-as-judge triangulation via Claude Opus 4.7 and gpt-4o. Per-field analysis via a dedicated script. All analysis scripts available at <https://github.com/Roverlucas/genai-reproducibility-protocol>

3. Data

Data availability statement. All 7,004 experimental records, provenance metadata (Run Cards in JSON format), W3C PROV-JSON provenance graphs, input abstracts (30 AI/ML + 10 PubMed PM2.5), HumanEval and GSM8K problem identifiers, and two-judge LLM-as-judge verdicts are deposited on Figshare with persistent DOI 10.6084/m9.figshare.31653373 (CC-BY 4.0). The Figshare deposit mirrors the project GitHub repository at <https://github.com/Roverlucas/genai-reproducibility-protocol> (release tag v1.1). Source data for all figures and tables are included. No restrictions on data availability apply.

4. Research involving human participants, their data, or biological material

Field	Response
Reporting on sex and gender	N/A — This study does not involve human participants. It evaluates the reproducibility of LLM API outputs using published scientific abstracts as input data.
Reporting on race, ethnicity, or other socially relevant groupings	N/A — No human participants.
Population characteristics	N/A — No human participants.
Recruitment	N/A — No human participants.
Ethics oversight	No ethics approval was required. The study uses only publicly available scientific abstracts and commercial/open-source LLM APIs. No human subjects, personal data, or sensitive information were involved.

5. Field-specific reporting

Select: **NONE** of the three options apply directly.

Our study is a computational experiment (software engineering / AI systems evaluation). It does not involve life sciences experiments, behavioural/social sciences with human subjects, or ecological/environmental fieldwork. If forced to choose, select “Life sciences” and fill the fields as follows:

If selecting “Life sciences study design”:

Field	Response
Sample size	Total 7,004 logged generative-AI runs across 9 deployment stacks and 6 task families: 30 AI/ML abstracts $\times$ 8 stacks $\times$ up to 4 tasks $\times$ up to 5 conditions $\times$ 5 reps; 30 HumanEval problems $\times$ 5 reps $\times$ 8 stacks; 30 GSM8K $\times$ 5 $\times$ 8; 10 PubMed PM2.5 $\times$ 5 $\times$ 8; multi-turn extension to gpt-4o and DeepSeek (50 runs each). Sample size determined to achieve power for detecting large effects (Cohen’s $d > 1.6$). Balanced 10-abstract subsample (Extended Data Table 4) confirms robustness.
Data exclusions	One Claude API run excluded due to API timeout returning empty output (49/50 retained). GPT-4 C1 summarisation limited to 3 abstracts (8 runs) due to API quota exhaustion. All exclusions documented in Supplementary Table S7 and noted in figure/table legends. No post-hoc exclusions of completed runs.
Replication	Every experimental condition replicated 5 times per item (or 3 times for temperature sweep conditions). Replication is the core subject of the study — reproducibility under fixed deterministic settings. All 7,004 runs available in the repository (Figshare DOI 10.6084/m9.figshare.31653373) for independent verification.
Randomization	Bootstrap CIs (10,000 resamples) quantify sampling uncertainty. Not applicable in the traditional sense. All stacks received identical inputs in identical order. Seed values for variable-seed condition (C2): {42, 123, 456, 789, 1024}. Two-judge LLM-as-judge sampling: 30 cases stratified across stack and disagreement kind.
Blinding	The two-judge LLM-as-judge protocol used blinded comparison: each judge scored pair (A, B) with stack/run identifiers withheld, against three pre-registered criteria (direction, magnitude $\pm 20\%$, CI overlap). For the rest of the study, blinding is not meaningful — automated metric computation across known stack identities.

6. Reporting for specific materials, systems and methods

Materials & experimental systems

Item	Status
Antibodies	n/a
Eukaryotic cell lines	n/a
Palaeontology and archaeology	n/a
Animals and other organisms	n/a
Clinical data	n/a
Dual use research of concern	n/a
Plants	n/a

Methods

Item	Status
ChIP-seq	n/a
Flow cytometry	n/a
MRI-based neuroimaging	n/a